

Rongfeng Cheng | Medical AI Research

Guangzhou, China

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Education

South China University of Technology (SCUT)

Bachelor of Management in Big Data Management and Application

Cumulative GPA: 3.46/4.0; average score: 85.18/100 (through Semester 6).

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Guangzhou, China

Sep 2023–Jun 2027

Selected Achievements & Deployed Work

Published: Co-authored a study on ethical safety in LLM code generation, published in *iScience* (2026).

IP: Two first-inventor patent applications and two software copyrights.

Clinical Deployment: Co-developed Dingbei Xinxin, deployed at Guangdong Second Provincial General Hospital; led the five-person Dingbei Meimei engineering and clinical-workflow team through its hospital launch; led the data-layer development for a health-management product, including a standardized schema, field governance, and medical terminology mapping for multi-source health-examination data.

Awards & Projects: Campus Gold, Internet+ Competition; Second Prize, Guangzhou Mathematical Modeling University League; Third Prize, CUMCM Guangdong Division; completed the SCUT SRP Student Research Training Project as Project Lead.

Research Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in

Jun 2026–Present

Deep Learning Models: An Application to 12-Lead ECG Classification

First author; revision in progress at IEEE JBHI

- **Problem:** Models can achieve strong discrimination while relying on evidence that conflicts with medical knowledge or becomes unstable under dataset shift.
- **Method:** Proposed and led an audit using domain-defined evidence units and matched controls, operator comparisons, and frozen-weight external transfer to test whether target effects exceed the strongest control and remain stable across datasets.
- **Finding:** Matched controls overturned the target-only 1AVB conclusion, operator choice changed AFIB and CRBBB verdicts, and external transfer exposed unstable evidence dependence; claims were classified as credible, limited, or unsupported.

DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification

Jun 2026–Present

First author; Manuscript in Preparation

- **Problem:** Nominally identical diagnostic labels can rely on different ECG evidence across centers; standard performance metrics reveal external degradation without explaining its source.
- **Method:** Proposed and led DLEF, which applies controlled perturbations along rhythm, conduction, QRS morphology, lead-specific QRS, and repolarization axes to form dataset-label fingerprints and directional source-target distances.
- **Finding:** Across three public datasets and six diagnoses, DLEF distance closely tracked label-level external AUROC degradation (Spearman $r = 0.843$) and outperformed prevalence, internal-performance, and embedding-shift baselines.

Internship & Product Leadership

AI Research Institute, Guangdong Second Provincial General Hospital

LLM Development Engineer Intern

Nov 2024–Present

- Developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

Guangzhou Yunzhan Zhichuang Information Technology Co., Ltd.

Backend & Algorithm Development Intern

Jul 2025–Jan 2026

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company's iTwins product.

Products & Team Projects

Dingbei Xinxin / CardioGPT: Cardiovascular Proactive Health LLM

2025–Present

Co-founder & Core Developer; Development Lead since Aug 2026

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

Myocardial Ischemia Knowledge Graph

2025

Project Lead, SCUT Student Research Program

Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

Publications & Manuscripts

Published [1]: J. He, S. Qin, H. Zou, et al., **R. Cheng**, H. Zhang, and W. Cheng. "CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation." *iScience*, 2026. doi:10.1016/j.isci.2026.115073.

Revised [2]: "Implicit Medical Ethical Deficits in Large Language Models: A Benchmarking and Alignment Study." *Nature Communications*; led the revision-stage experiments for the response to reviewers.

Revision in Progress [3]: **R. Cheng et al.** "A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification." *IEEE JBHI*. First author.

In Preparation [4]: **R. Cheng et al.** "DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification." *Manuscript in Preparation*. First author.

Ongoing: PathCLIP: Evidence-Aware ECG–Text Representation Learning. ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

Skills

Programming & Frameworks: Python, PyTorch, scikit-learn, TypeScript, SQL

Modeling Methods: XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

LLM & Knowledge Systems: LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

Evaluation & Data Curation: RAG/retrieval evaluation, perturbation analysis, external validation, schema design

Engineering: Neo4j, Next.js, Flask, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528